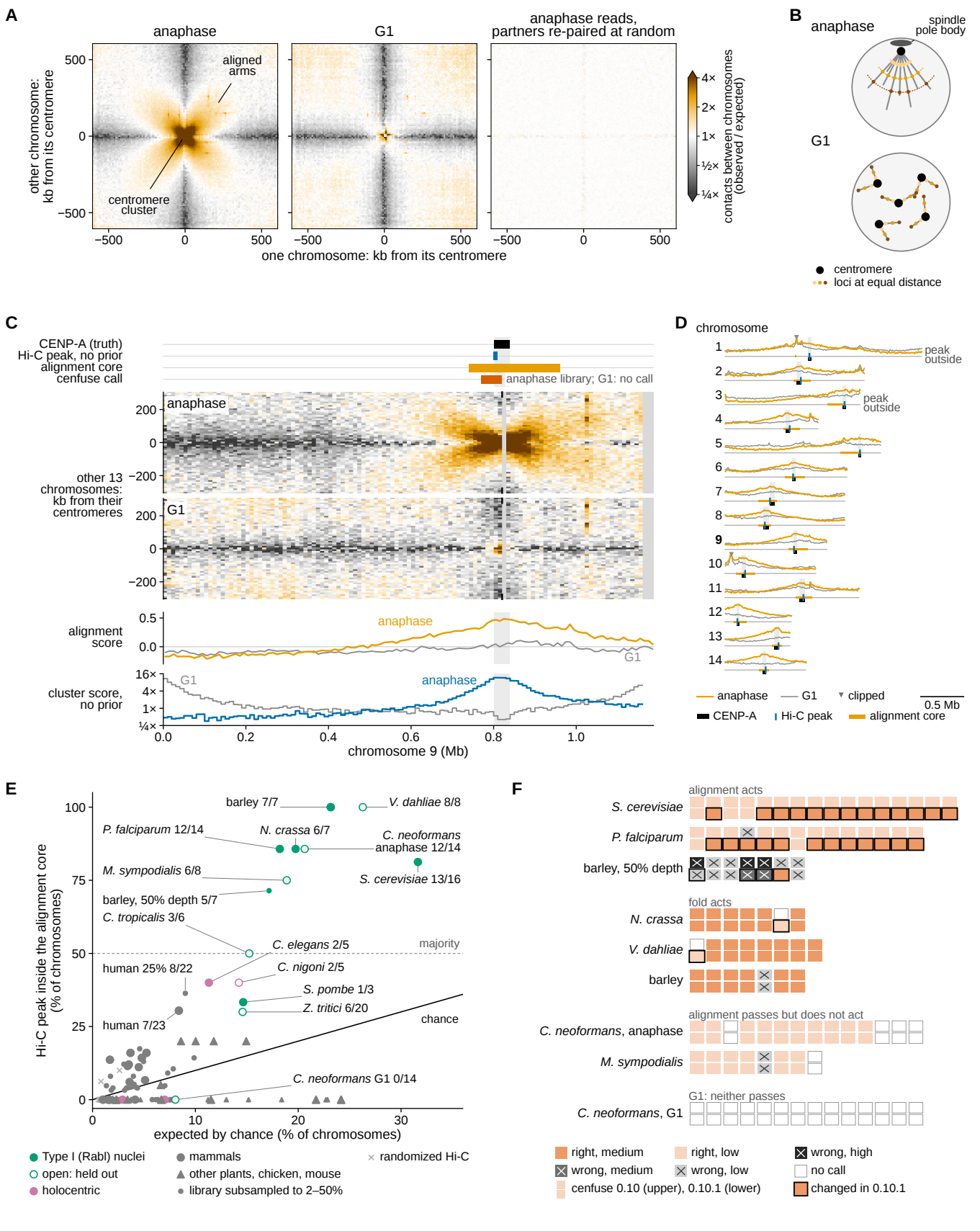

A
anaphase
G1
anaphase reads, partners re-paired at random
aligned arms
centromere cluster
other chromosome: kb from its centromere
one chromosome: kb from its centromere
contacts between chromosomes (observed / expected)
B
anaphase
spindle pole body
G1
centromere
loci at equal distance
C
CENP-A (truth)
Hi-C peak, no prior
alignment core
cenfuse call
anaphase library; G1: no call
anaphase
G1
other 13 chromosomes: kb from their centromeres
alignment score
cluster score, no prior
chromosome 9 (Mb)
D
chromosome
peak outside
anaphase
G1
clipped
0.5 Mb
CENP-A
Hi-C peak
alignment core
E
Hi-C peak inside the alignment core (% of chromosomes)
expected by chance (% of chromosomes)
barley 7/7
V. dahliae 8/8
P. falciparum 12/14
N. crassa 6/7
C. neoformans anaphase 12/14
M. sympodialis 6/8
S. cerevisiae 13/16
barley, 50% depth 5/7
C. tropicalis 3/6
C. elegans 2/5
majority
human 25% 8/22
C. nigoni 2/5
S. pombe 1/3
Z. tritici 6/20
human 7/23
chance
C. neoformans G1 0/14
Type I (Rabl) nuclei
open: held out
holocentric
mammals
other plants, chicken, mouse
library subsampled to 2–50%
randomized Hi-C
F
alignment acts
S. cerevisiae
P. falciparum
barley, 50% depth
fold acts
N. crassa
V. dahliae
barley
alignment passes but does not act
C. neoformans, anaphase
M. sympodialis
G1: neither passes
C. neoformans, G1
right, medium
right, low
wrong, high
wrong, medium
wrong, low
no call
cenfuse 0.10 (upper), 0.10.1 (lower)
changed in 0.10.1

In nuclei that hold their centromeres at one pole, chromosome arms lie side by side by distance from the centromere, so contacts alone mark centromeres. Seen in fungi, a malaria parasite and barley, it fades as *Cryptococcus* is held in G1 and is weak or absent in territorial and holocentric nuclei.

A, Contacts between chromosomes, every pair stacked on its two centromeres. Anaphase *Cryptococcus neoformans* nuclei draw an X through the centromere cluster: loci equally far from their own centromeres touch. G1 nuclei of the same strain barely do; the same reads re-paired at random do not.

B, A Rabl-like polarity: centromeres held at the spindle pole and arms trailing behind put equally distant loci side by side; in G1 the centromeres scatter.

C, One typical chromosome against all others. In anaphase the X sits on its centromere, and the alignment score, which reads that X at every position, peaks on the CENP-A domain, as does the centromere-cluster score. In G1 neither does.

D, All 14 chromosomes: in anaphase the alignment peaks at the centromere on nearly every one; in G1 it barely rises.

E, Every Hi-C library tested: only nuclei with a Rabl-like polarity place the contact-only centromere inside the arm alignment beyond chance and on most chromosomes.

F, What the centromere caller does with it: right calls gain confidence or move onto the centromere; no wrong call gains any.

Terms. Contacts are balanced and shown as observed/expected (O/E) within each pair of chromosomes. *Hi-C peak, no prior*: the summit of cenfuse's centromere-cluster score, found without sequence or position priors. *Alignment score* of a position: the mean contact excess (O/E above each locus's own mean) between the chromosome's loci and loci of the other chromosomes as far from their centromeres as the locus is from that position; *core*: positions above half its peak. A chromosome *agrees* when its prior-free peak touches the core; chance = (core + 2 peak lengths) / scanned length; a genome passes at Poisson-binomial $P \le 0.005$ with a majority agreeing (at least 3 tested). *Type I* nuclei: centromeres clustered, Rabl-like (Hoencamp et al. 2021). *C. neoformans* H99: CENP-A ChIP truth (Yadav et al. 2018); Hi-C of one strain (GAL7p-MPS1, TUB1-GFP), synchronized in anaphase and arrested in G1 by Polisetty et al. (2025), 20.4 and 20.1 million read pairs, 10-kb bins.

A, Mean O/E over all 182 ordered pairs of the 14 chromosomes, each locus placed by its signed distance from its own CENP-A midpoint. Randomized: the anaphase reads with partners re-paired at random, coverage kept. Anaphase: centromere cluster 15.5× (within 20 kb of both centromeres); 50–200 kb out, contacts on the X are 1.85× those 150 kb beside it. G1: 3.4× and 1.09×; distal arm ends meet instead (corners), and all 14 G1 prior-free peaks lie at a chromosome end. Randomized: 1.0× and 1.00×.

B, Schematic, not data. Anaphase: centromeres clustered at the spindle pole body, arms trailing (Rabl-like polarity); *C. neoformans* kinetochores cluster from late G2 into mitosis and scatter in G1 (Polisetty et al. 2025).

C, Chromosome 9 (CP003828.1, 1.19 Mb), the lower median (rank 7 of 14) of cenfuse's anaphase alignment heights, not the most striking. Top: CENP-A at 802–839 kb (gray band; ticks on the maps) and cenfuse's results on the anaphase library. Maps: contacts with the 13 other chromosomes, each stacked on its CENP-A midpoint (scale as in A; vertical axis compressed; gray, unmappable). Alignment score with partners placed on CENP-A in both states; in anaphase cenfuse's own score, placed on the prior-free peaks, is the same ($r \ge 0.98$ on 13 of 14 chromosomes). Bottom: cluster score, × chromosome median.

D, All chromosomes to scale, scores as in C on one axis (▾, clipped above 0.65). In anaphase the core holds the prior-free peak on 12 of 14 (2.9 expected, $P = 1.5\times10^{-7}$); on chromosome 1 a narrow spike, present in both states, takes the maximum; on chromosome 3 the peak lies 10 kb beyond the core. Placed on CENP-A, G1 keeps a weak alignment (CENP-A in the core on 8/14, $P = 0.0014$; anaphase 12/14, $P = 2.0\times10^{-6}$), but with no centromere cluster to anchor the partners the prior-free test finds 0/14.

E, All 159 axis tests of the benchmark (9 randomized runs with too few prior-free peaks are untested, not drawn); line, chance; dashed, majority; open symbols, genomes held out until the rules were frozen. Passing: *S. cerevisiae* 13/16 ($P = 3.4\times10^{-5}$), *P. falciparum* 12/14 (3.6×10^{-8}), *N. crassa* 6/7 (3.1×10^{-4}), barley 7/7 (3.4×10^{-5}; two runs, one library); held out, *V. dahliae* 8/8 (2.1×10^{-5}), *C. neoformans* 12/14 (1.5×10^{-7}), *M. sympodialis* 6/8 (4.1×10^{-4}; truth from the kinetochore protein Mtw1, not CENP-A), barley at 50% depth 5/7 (0.0020). None of 46 randomized, 23 mammalian, 19 other plant, chicken and mouse, or 4 holocentric runs passes, nor any of 55 libraries subsampled to 2–50% but barley at 50%; in human a minority agrees beyond chance (7/23, $P = 0.0013$; 8/22, $P = 3.1\times10^{-4}$, at 25% depth). Type I exceptions: *Z. tritici* 6/20 ($P = 0.055$; 8 accessory chromosomes), *C. tropicalis* 3/6 (0.049; 3C-seq, truth placed by sequence), *S. pombe* 1/3. *C. elegans* 2/5 ($P = 0.095$) gave 3/5 ($P = 0.0069$) before the centromere cluster was left out of the score. Same-library runs coincide.

F, Every chromosome of the runs where a Rabl-axis test passed, plus G1, scored against truth (within one Hi-C bin; barley four). The alignment acts only where the arm fold, the cis readout of the same axis, leans the same way ($P \le 0.05$): 13 *S. cerevisiae* and 11 *P. falciparum* right calls rise from low to medium, *P. falciparum* chromosome 4 and one barley chromosome at 50% depth move onto their centromeres, and three high-confidence wrong barley calls are lowered. Where the fold passes (*N. crassa*, *V. dahliae*) a right low call is added. In *C. neoformans* and *M. sympodialis* the fold has no power (1/14, $P = 0.71$; 0/8, $P = 1$), so their right calls (10/14, 6/8) stay low. Confidently wrong calls added: 0.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1 · Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/figures/benchmark_hic2/benchmark_hic2.py · 2026-11-03